

Sarah Cruz

Pace University CS 676 Project #1

Credibility Assessment

I. Introduction

The credibility of the sources used in a chatbot's responses is imperative to ensure the trustworthiness and accuracy of information. Many individuals use chatbots to obtain their information. If this information comes from sources with varying levels of credibility, it becomes a major issue. The focus of this project is assessing the credibility of the sources used for research in chatbots. A source is scored on a scale from 0-1, with zero representing low credibility and one representing high credibility.

The original algorithm focused mainly on the URL, which led to recognition of certain domains. A weakness was that the webpage was not being examined. This led to difficulty in evaluating unrecognized websites.

The main goal of this project was to improve the provided credibility-scoring algorithm. Rather than focusing heavily on the URL, it was built to gather information from the webpage for unrecognized domains. Additionally, distinguishing between varying types of sources, including preprints and peer-reviewed sources, was targeted for improvement.

II. Algorithm and Rationale

The provided algorithm used rule-based scoring. Credibility was scored on a scale of 0-1. To increase the accuracy of the credibility assessment, webpage-level information was added, particularly for unfamiliar domains. This approach was chosen because the original algorithm did not read the webpage information as part of its credibility assessment. The changes in the algorithm are categorized by webpage retrieval, author, publication date, references, corrections policy, DOI, preprint and peer-reviewed sources, URL-path penalties,

and an optional LLM layer. Out of the improvement opportunities offered for this project, this algorithm focused on the “read the page” improvement, which included looking at the webpage in addition to the URL.

The rule-based score provides a baseline level of credibility. When the algorithm recognizes the domain, it uses the credibility value associated with it. Instead of completely removing this feature, there were additional credibility signals added to enhance the rule-based scoring.

For webpage retrieval, the code `_page_credibility_signals()` was added. This was used to attempt to obtain the webpage from unrecognized domains, while recognized domains used their current domain information. This pushed the algorithm to look beyond the URL. For the author portion, it was coded to add +0.05 when an author was identified. The rationale behind this was to find more detail about the source rather than the URL alone. Author information and expertise were identified as components of web credibility assessment (Choi & Stvilia, 2015). For publication date, +0.04 was added when it was detected. This provides an extra layer of information about the webpage. For references, +0.06 was added when references or citations were found on the webpage. This provided additional positive credibility signals. When terms mentioning a corrections policy were found, +0.05 was added. The rationale behind this was that the presence of a corrections policy represents greater accountability and transparency, which were identified as relevant measures in assessing web credibility (Choi & Stvilia, 2015). For DOI, +0.08 was added for webpage DOI metadata. This expanded the algorithm’s original DOI scoring.

URL-path penalties were used to potentially determine the type of content based on the URL. For example, `/forum/`, `/sponsored/`, `/opinion/`, and `/blog/`. This is useful for the chatbot to determine what type of content they are evaluating and whether the content has credible characteristics (Choi & Stvilia, 2015). For preprints, -0.20 was applied when a preprint signal was detected. Additionally, recognized domains were differentiated to distinguish between

preprint sources and peer-reviewed publications. This was done because preprints do not go through the same process as peer-reviewed publications (Carneiro et al., 2020). Rather than assuming a preprint was unreliable, the goal was for the algorithm to gain the ability to differentiate between those two types of sources.

Each individual signal weight was chosen manually. Webpages with positive signals were adjusted in small amounts since there was not a single characteristic that was intended to have a larger influence independently. The magnitude for each adjustment was determined by its intended relative impact in the algorithm, while considering the scores presented by the original code.

Once the applicable credibility signals on a given webpage were detected, the positive and negative adjustments were applied to the original credibility score. This was used to calculate the rule-based score. The rule-based score was kept within the 0-1 range.

After the calculation of the rule-based score, Claude was used to assist in determining the credibility. When Claude is available, the final credibility score is 60% rule-based and 40% Claude assessment. This weight distribution was chosen because the rule-based portion was kept as the primary influence, and the Claude portion was used for additional judgment that may not have been found in the rules. Recent research has been examining the growing use of LLMs in automatic credibility assessment (Srba et al., 2026). This allows both the credibility signals and LLM judgment to contribute to the final score. It is used as a supplemental component if available and does not replace the rule-based score. If Claude is unavailable, the algorithm will use only the rule-based score.

A failure handling strategy was also included in the algorithm with the goal of preventing the scorer from crashing if a webpage could not be retrieved. A five-second timeout was included for webpage retrieval. A webpage request may fail if an error occurs or the webpage is unavailable. If retrieval fails, the code `_page_credibility_signals()` does not return webpage

signals, and the algorithm continues its assessment using the remaining rule-based signals. Similarly, if Claude is unavailable, the algorithm uses only the rule-based credibility score.

III. Literature Review

Credibility is considered a multifaceted construct that cannot be assessed by focusing on a single characteristic. According to Choi and Stvilia (2015), key dimensions of credibility include trustworthiness, expertise, dynamism, goodwill, objectivity, and personal attraction. Operator trustworthiness is measured by commercial implication (URL endings, ads), integrity (sponsored vs. nonsponsored links), transparency (notice of completed corrections, author information), and decency (the author's tone). The credibility dimension of expertise can be defined by the author's credentials, previous publications, and affiliations. Additionally, credibility can be evaluated by the characteristics of the webpage, such as recency and neutral/unbiased information. Overall, this demonstrates how credibility is dependent on multiple factors.

The authors Schwarz and Morris (2011) shed light on the danger of users trusting incorrect or misleading content from web sources. They explored the webpage features that could help provide greater accuracy in credibility assessments. They created a dataset including 1,000 webpages across the five topics of health, finance, politics, celebrity news, and environmental science. They were scored on their credibility using a five-point Likert scale. Three categories of features were explored, including on-page features, off-page features, and aggregate features. On-page features include spelling errors, the type of domain (.gov, .com), and advertising. Off-page features included supplementary data about the website, including awards, PageRank, and the amount of sharing. Aggregate features were determined by one month of anonymous browsing data. This included overall popularity, revisitation patterns, their geographic reach, and how long a user kept the URL open versus the amount of time they viewed the page. It was found that the individual features mentioned above were not strong predictors of credibility. The highest correlations were $p = .3-.5$, which represented moderate

correlations. However, it was concluded that these findings still provide useful information about the signals that could have an impact on credibility assessment.

After these initial features were examined, the researchers conducted an experiment where selected features were displayed in a visual format. The researchers showed the participants additional information to examine with the search results. The augmented search results included awards/certifications, popularity among topic experts, geographic reach, and PageRank. These augmented search results increased the accuracy of participants' credibility assessments (Schwarz & Morris, 2011). This finding supports the use of providing a credibility score along with its rationale for why the source received that score.

Carneiro et al. (2020) analyzed the differences between the quality of reporting in preprints and peer-reviewed articles. Preprints are described as “complete manuscripts submitted to publicly accessible repositories”, with the potential of being further submitted to formal scientific journals. Peer review is the process of researchers from relevant fields reviewing the articles to determine their adequacy or quality for publication. Researchers compared bioRxiv preprints and PubMed peer-reviewed articles. Additionally, preprints were also compared with their own later peer-reviewed versions. It was found that peer-reviewed articles did receive higher overall reporting scores, which were measured by the percentage of applicable criteria provided by reporting guidelines and a checklist; however, the difference was small (67.6 for preprints and 72.3 after peer review). This shows that preprints are not necessarily unreliable; rather, being able to differentiate them is crucial in the overall algorithm when assessing the article's credibility on a case-by-case basis.

In the literature review written by Metzger (2007), the author evaluated methods used to assess the credibility of online sources. Five criteria were used to assess credibility, including accuracy, authority, objectivity, currency, and coverage/scope. Authority can be determined by contact information provided for the organization and whether the author's credentials and qualifications are present. Objectivity is meant to find the underlying purpose of the site, whether

it is fact or opinion-based. Currency assesses whether the information is up to date by exploring its publication/update dates. Additionally, the author suggests that signals such as the absence of advertising, scientific references, domain/URL, editorial review, and source citations are factors in credibility assessment. These findings support the algorithm's approach in examining multiple characteristics of a webpage rather than just the URL.

A systematic literature review by Srba et al. (2026) examined automatic credibility assessment and credibility signals in the era of large language models (LLMs). This study reviewed 175 Natural Language Processing (NLP) studies about automatic credibility assessment and credibility signal detection. The authors found that credibility assessment involves both detecting credibility signals and combining those signals into an overall evaluation. Credibility signals assessed include bias, persuasion techniques, text quality, logical fallacies, and factuality. These signals can then be combined, and an automated system assigns them to a credibility category or numerical score. Some algorithms use different methods, ranging from rule-based to advanced AI systems. These advanced systems can provide more nuanced and context-aware analysis. The authors discovered that in newer credibility assessment research, the use of LLMs has increased, although their full potential has not yet been explored. This relates to the improved algorithm because it combines rule-based methods with an LLM as a supplemental feature.

Overall, the literature supports multiple components when assessing credibility, rather than relying heavily on the URL. These characteristics include references, domain information, author information, publication date, and corrections when evaluating different sources. This information supports the changes made to the original algorithm, as it includes multiple webpage-level approaches along with preprints and peer-reviewed sources. Additionally, the literature shared that there was potential for using advanced AI as supplemental assistance when doing automatic credibility assessment. In conclusion, these findings support the adjustments that were made to improve the original credibility-scoring algorithm.

IV. Quantitative Evaluation and Results

The performance of the adjusted credibility algorithm was evaluated using `evaluate.py`. This evaluation included 24 URLs and compared the starter and improved algorithms using the rules-only and LLM-assisted conditions. The three metrics used for this evaluation were the mean absolute error (MAE), band accuracy, and worst single error. MAE measures the average absolute difference between predicted and expected values. Band accuracy measures the percentage of predicted credibility scores that were in the correct credibility category. The worst single error is the largest difference between the predicted and expected credibility scores.

Version	MAE	Band Accuracy	Worst Single Error
Starter algorithm - rules only	0.142	66.7%	0.410
Starter algorithm - rules + LLM	0.086	83.3%	0.230
Improved rule-based algorithm	0.053	91.7%	0.170
Improved rule-based algorithm + LLM	0.047	91.7%	0.140

For the rules-only comparison, the MAE decreased from 0.142 to 0.053. For the LLM-assisted comparison, the MAE decreased from 0.086 in the starter algorithm to 0.047 in the improved algorithm. The rule-only band accuracy increased from 66.7% to 91.7%, while the LLM-assisted band accuracy increased from 83.3% to 91.7%. The worst single error decreased from 0.410 to 0.170 for the rule-based comparison and from 0.230 to 0.140 with LLM assistance. This demonstrates that the improved rule-based algorithm outperformed the baseline algorithm on all three evaluation metrics. The addition of Claude to the improved algorithm made slight improvements to MAE and worst single error; however, it did not change the band accuracy.

In addition to the evaluation metrics, URL results were examined using expected score, predicted score (labeled as “got” in `evaluate.py`), and error. These results showed the difference between the algorithm’s predicted and expected credibility scores for each URL. For example,

the Sponsored Miracle-Supplement URL had an expected score of 0.10, a predicted score of 0.17, and an error of 0.07. This evaluation described this source as “sponsored commercial content.” The largest error in the improved rule-based algorithm happened in the Seeking Alpha URL. The expected score was 0.35, and the predicted score was 0.52, resulting in an error of 0.17. These results show that although overall error decreased, differences remained between predicted and expected URL scores, demonstrating remaining prediction error.

V. Limitations and Remaining Failures

Although the algorithm demonstrated measurable improvements, there are still weaknesses from the baseline algorithm that were not changed. Since the webpage retrieval in this project is focused on unrecognized domains, known domains rely on their previously assigned credibility scores. These known domains are not evaluated using the same webpage-level analysis as unknown domains. Another weakness is that the signal weights were created manually rather than being optimized statistically. Retractions and uncertainty/confidence estimates were not included in this algorithm, which could be addressed further. Individual errors still occur for some URLs, as demonstrated with the Seeking Alpha URL.

VI. Conclusion

The goal of this project was to improve the provided credibility-scoring algorithm by expanding the amount of information used to assess online resources. Webpage-level credibility signals and a Claude layer were added, with the rule-based approach as the main component. Improvements were seen in all three evaluation metrics: MAE went from 0.142 to 0.053 for the rules-only comparison and from 0.086 to 0.047 for the LLM comparison. Band accuracy went from 66.7% to 91.7% for rules only and from 83.3% to 91.7% with the LLM. Worst single error went from 0.410 to 0.170 for rules only and from 0.230 to 0.140 with the LLM. In conclusion, the

results showed that providing the algorithm with additional information, including author information, references, correction policies, DOI information, and publication date, can improve credibility assessment.

References

- Carneiro, C. F. D., Queiroz, V. G. S., Moulin, T. C., Carvalho, C. A. M., Haas, C. B., Rayêe, D., Henshall, D. E., De-Souza, E. A., Amorim, F. E., Boos, F. Z., Guercio, G. D., Costa, I. R., Hajdu, K. L., van Egmond, L., Modrák, M., Tan, P. B., Abdill, R. J., Burgess, S. J., Guerra, S. F. S., . . . Amaral, O. B. (2020). Comparing quality of reporting between preprints and peer-reviewed articles in the biomedical literature. *Research Integrity and Peer Review*, 5, Article 16. <https://doi.org/10.1186/s41073-020-00101-3>
- Choi, W., & Stvilia, B. (2015). Web credibility assessment: Conceptualization, operationalization, variability, and models. *Journal of the Association for Information Science and Technology*, 66(12), 2399–2414. <https://doi.org/10.1002/asi.23543>
- Metzger, M. J. (2007). Making sense of credibility on the Web: Models for evaluating online information and recommendations for future research. *Journal of the American Society for Information Science and Technology*, 58(13), 2078–2091. <https://doi.org/10.1002/asi.20672>
- Schwarz, J., & Morris, M. R. (2011). Augmenting web pages and search results to support credibility assessment. In *Proceedings of the SIGCHI Conference on Human Factors in Computing Systems* (pp. 1245–1254). Association for Computing Machinery. <https://doi.org/10.1145/1978942.1979127>
- Srba, I., Razuvaevskaya, O., Leite, J. A., Moro, R., Schlicht, I. B., Tonelli, S., Moreno García, F., Barrio Lottmann, S., Teyssou, D., Porcellini, V., Scarton, C., Bontcheva, K., & Bielikova, M. (2026). A survey on automatic credibility assessment using textual credibility signals in the era of large language models. *ACM Transactions on Intelligent Systems and Technology*, 17(2), Article 26. <https://doi.org/10.1145/3770077>